

Network Security

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ES ZG513/ SS ZG513, Network Security
Lecture No VIII.

Learning Objectives

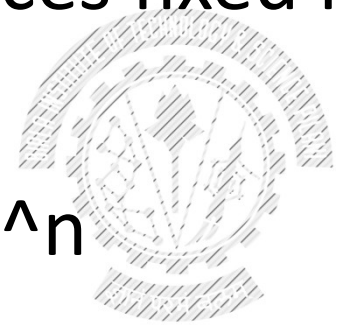
- Understand cryptographic hash functions
- Learn properties of secure hash functions
- Study SHA-512 algorithm structure
- Understand Message Authentication Codes (MAC)
- Learn HMAC and CMAC algorithms
- Work through practical examples

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Cryptographic Hash Function

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- A hash function H takes input of arbitrary length and produces fixed length output.

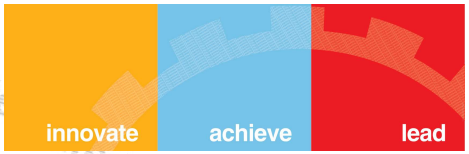


- H : {0,1}* → {0,1}^n

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- H(m) is called digest.

Important Properties of Hash Functions



1. Preimage resistance
2. Second preimage resistance
3. Collision resistance
4. Deterministic output
5. Efficient computation
6. Avalanche effect

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Preimage Resistance

- Given y = H(m), it should be computationally infeasible to find m.

- Formal:
- Given y, find m such that H(m)=y

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Second Preimage Resistance

- Given m1, it should be hard to find m2 ≠ m1 such that:

- H(m1) = H(m2)



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Collision Resistance

- It should be computationally infeasible to find ANY two messages:
- $m1 \neq m2$ such that
- $H(m1) = H(m2)$

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Avalanche Effect

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- A small change in input causes large change in output.

- Example:
- H('HELLO') ≠ close to H('HELLo')

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Applications of Hash Functions

- Password storage
- Digital signatures
- Message authentication
- Blockchain
- File integrity verification

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Secure Hash Algorithm Family

- Developed by NIST

- SHA-1 (deprecated)
- SHA-2 family:
- SHA-224
- SHA-256
- SHA-384
- SHA-512

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SHA-512 Overview

- Output length: 512 bits
- Block size: 1024 bits
- Word size: 64 bits
- Number of rounds: 80

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SHA-512 High Level Steps

1. Padding the message
2. Append message length
3. Initialize hash values
4. Process blocks using compression function
5. Produce 512-bit hash output

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SHA-512 Padding

- Message is padded so that:
- Length ≡ 896 mod 1024
- Padding includes:
- 1 bit '1' followed by zeros
- Then 128-bit length field

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SHA-512 Compression Function

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- For each block:

- Message schedule expansion
- 80 rounds of mixing
- Logical functions:
- Ch, Maj, Σ0, Σ1

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Example (Conceptual) SHA-512

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- Message: 'ABC'

- Step 1: Convert to binary
- Step 2: Apply padding
- Step 3: Process 1024-bit block
- Step 4: Final 512-bit digest

- Actual output example:
- ddaf35a193617abacc417349ae204131...

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Message Authentication Codes

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- A MAC ensures:
- Message integrity
- Authentication

- MAC_k(m) → tag

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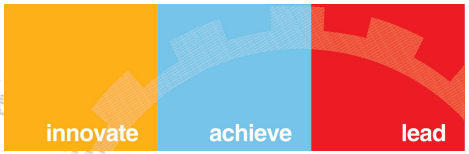
MAC Security Goal

Adversary should not be able to forge a valid tag for a new message even after seeing many message-tag pairs.



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Types of MAC Constructions



1. Hash based MAC (HMAC)
2. Block cipher based MAC (CMAC)
3. CBC-MAC

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HMAC Overview

- HMAC uses a hash function and secret key.

- HMAC(K,m) = H((K $\oplus$ opad) || H((K $\oplus$ ipad) || m))

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HMAC Parameters

- ipad = inner padding
- opad = outer padding

Typical hash functions used:

- SHA-256
- SHA-512

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Worked Example: HMAC (Toy)

- Let:
- K = 101010
- m = 110011

- Step 1: K ⊕ ipad
- Step 2: Hash with message
- Step 3: K ⊕ opad
- Step 4: Final hash → MAC

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CMAC Overview

- CMAC is based on block ciphers.
- Uses AES encryption to generate authentication tag.

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CMAC Steps

1. Generate subkeys K1 and K2
2. Divide message into blocks
3. Apply CBC-MAC structure
4. Encrypt final block
5. Output authentication tag

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CMAC Worked Example (Conceptual)

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- Message blocks: M1, M2, M3

- C1 = AES_K(M1)
- C2 = AES_K(M2 $\oplus$ C1)
- C3 = AES_K(M3 $\oplus$ C2)

- Tag = C3

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Comparison: HMAC vs CMAC

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HMAC:

- Based on hash functions
- Widely used in internet protocols

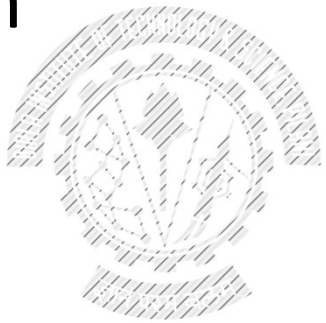
CMAC:

- Based on block ciphers
- Efficient when AES hardware exists

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Real World Uses

- TLS authentication
- API authentication
- Secure messaging
- Banking systems



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Summary

- Hash functions provide data integrity
- SHA-512 is a secure hash algorithm in SHA-2 family
- MAC provides authentication and integrity
- HMAC uses hash functions
- CMAC uses block ciphers

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